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AI-BASED SMART RETAIL ANALYTICS & Inventory Management

DeepLearningfortheModernStore–MLP,OptimizerAnalysis,YOLOVision,LSTM Forecasting & GAN Simulation

MLP | Optimizers | YOLO | LSTM | GAN

Course Project Presentation

THE CHALLENGE

Retail Runs on Guesswork Without AI

Physicalandonline retailersgeneratemassivestreamsoftransaction,footfall,andshelfdataevery day – yet most decisions on stocking, staffing, and personalization are still made manually.

Stockouts & Overstocking

No predictive model to forecast demand leads to lost sales or wasted capital tied up in excess inventory.

Unknown Customer Behavior

Purchase history sits unused – retailers can't predict what a shopper is likely to buy next.

Blind Store Floors

No automated way to count footfall or detect empty shelves in real time.

Sparse Movement Data

Limited in-store movement data weakens recommendation engines and layout planning.

Goal: Build an end-to-end AI pipeline that senses, predicts, and simulates retail behavior to optimize inventory and customer experience.

PROJECT ROADMAP

Five AI Modules, One Retail Pipeline



Camera & POS data flow left to right – from real-time in-store sensing to demand forecasting and simulation that feeds a smarter recommendation engine.

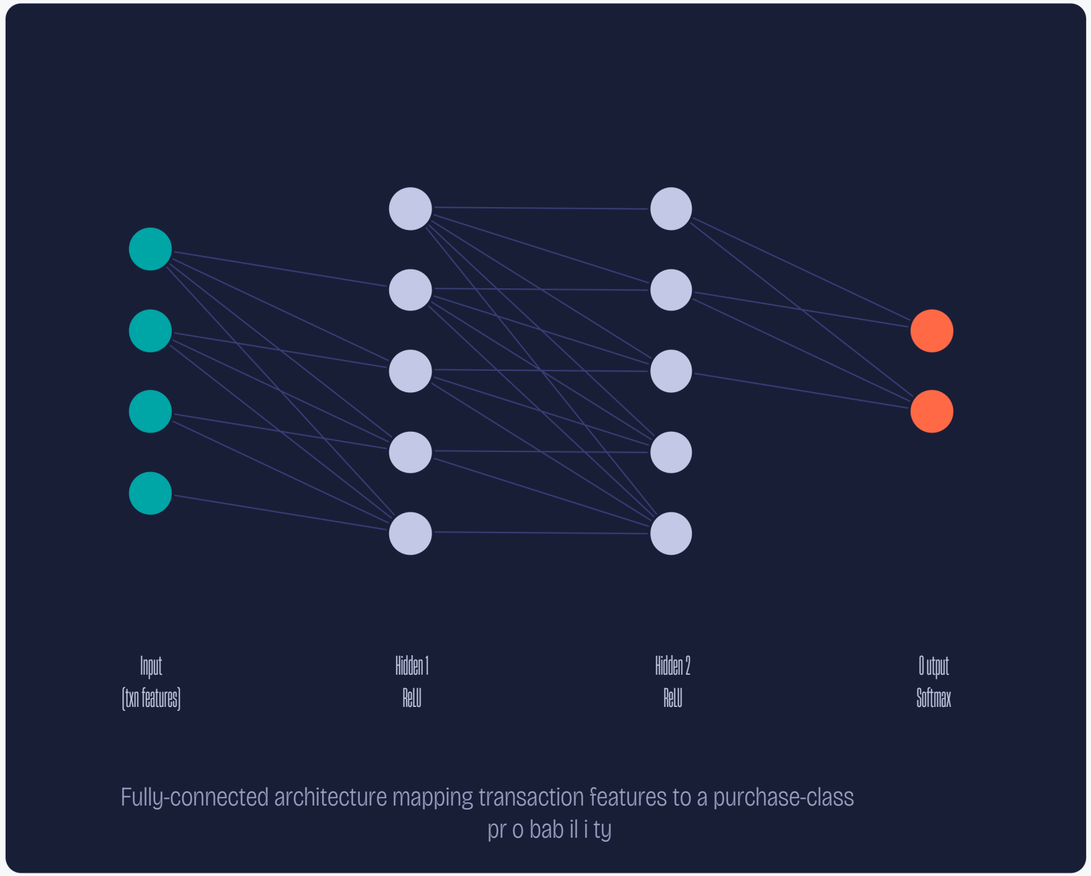
5 AI Models Integrated

3 Data Sources: POS, Camera, History

1 Unified Analytics Dashboard

MODULE 1

Predicting Purchasing Behavior with an MLP



Fully-connected architecture mapping transaction features to a purchase-class probability

How it works

Input Features

Transaction history: purchase frequency, basket value, category mix, recency, time-of-day patterns.

Hidden Layers

Two dense layers with ReLU activation and dropout learn non-linear purchase patterns.

Output

Softmax layer predicts purchase-class probability (e.g., will buy / high-value / churn-risk).

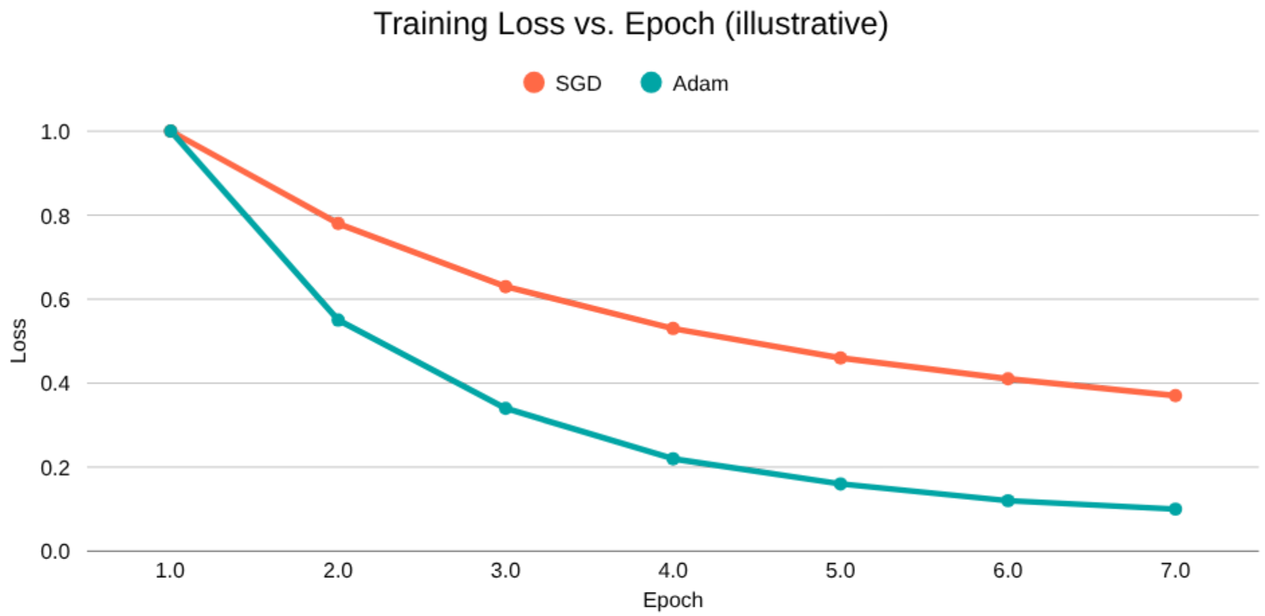
Why an MLP

Fast to train on tabular transaction data and provides an interpretable baseline before deepsequence models.

MODULE 2

SGD vs. Adam: Choosing the Right Optimizer

Benchmarkedforcustomerclassification(MLP)andsales-predictiontaskstoselectthefastest,moststabletraining strategy.



SGD

- Simple, low memory footprint
- Needs careful learning-rate tuning
- Slower, steadier convergence
- Can generalize better with momentum

Adam

-Adaptive per-parameter learning rates

- Fastconvergence, fewer epochsneeded
- Robustdefaultfor sparse transactiondata

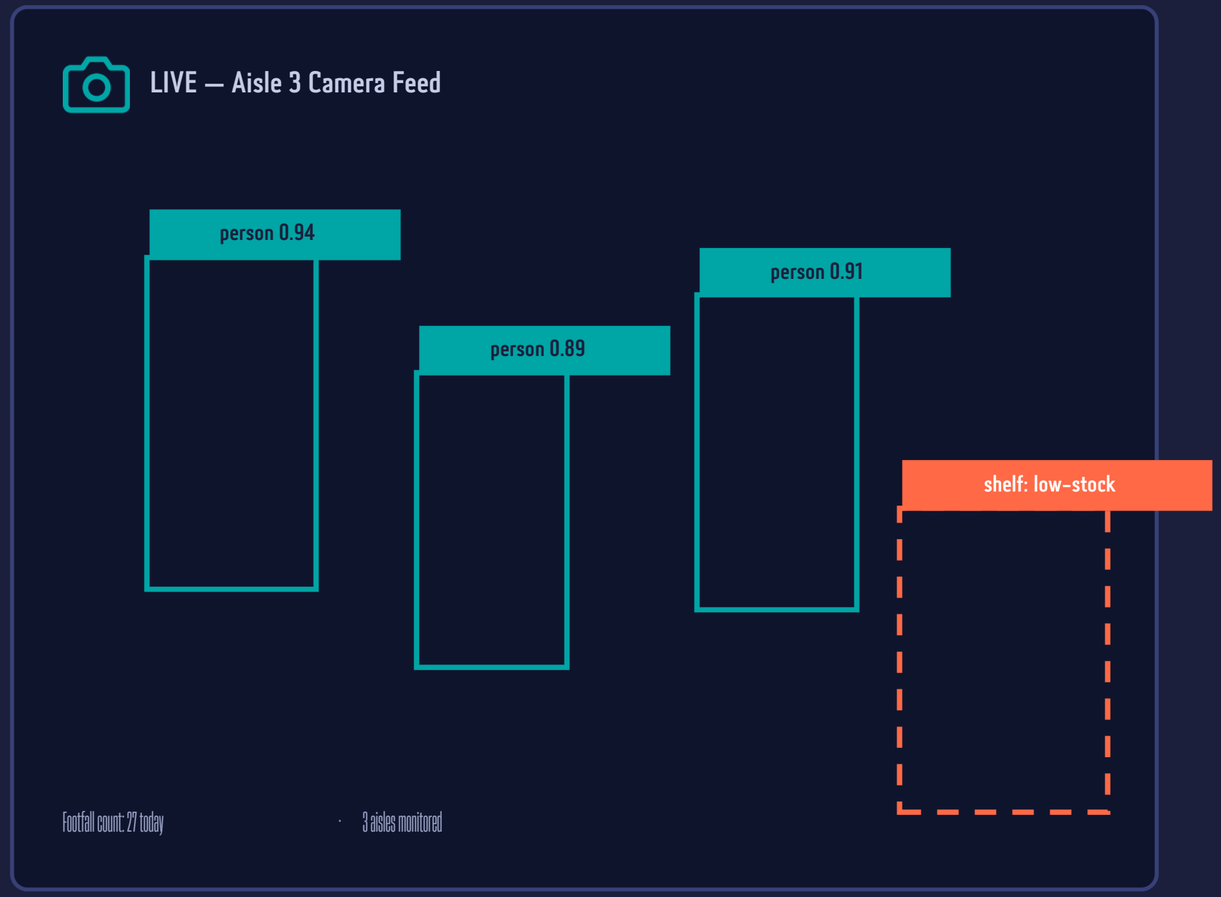
-Selected as primary optimizer for this project

Result

Adam converges roughly 2x faster than SGD on both the MLP classifier and sales-prediction regressor, making it the default optimizer – while SGD with momentum remains a strong, lightweight fallback for production retraining.

MODULE 3

YOLO: Real-Time Eyes on the Store Floor



Two jobs, one detector

Customer Counting

YOLO detects and tracks people frame-by-frame to measure real-time footfall and dwell timeperzone.

Shelf Monitoring

The same detector recognizes product-shelf regions to flag gaps and low-stock conditions instantly.

Why YOLO

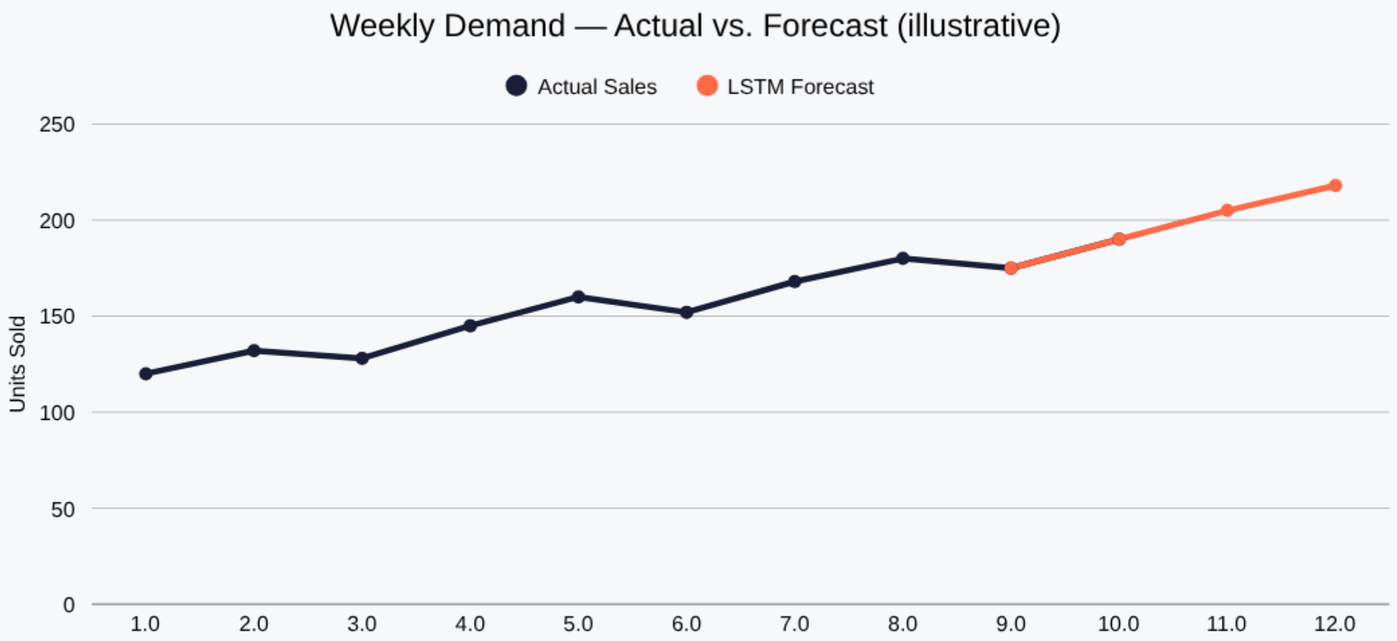
Single-pass detection gives real-time speed (30+ FPS) suitable for continuous in-store video streams.

Feeds the Pipeline

Counts and stock alerts feed directly into the demand-forecasting and inventory-alert systems.

LSTM: Forecasting Product Demand

A recurrent network learns seasonal and trend patterns from historical sales sequences to forecast future demand per SKU.



Sequential Memory

LSTM cells retain long-range trend and seasonality signals that simple models miss.

I npu t

Weekly/daily sales history per product, price, promotions, and holiday flags.

Output

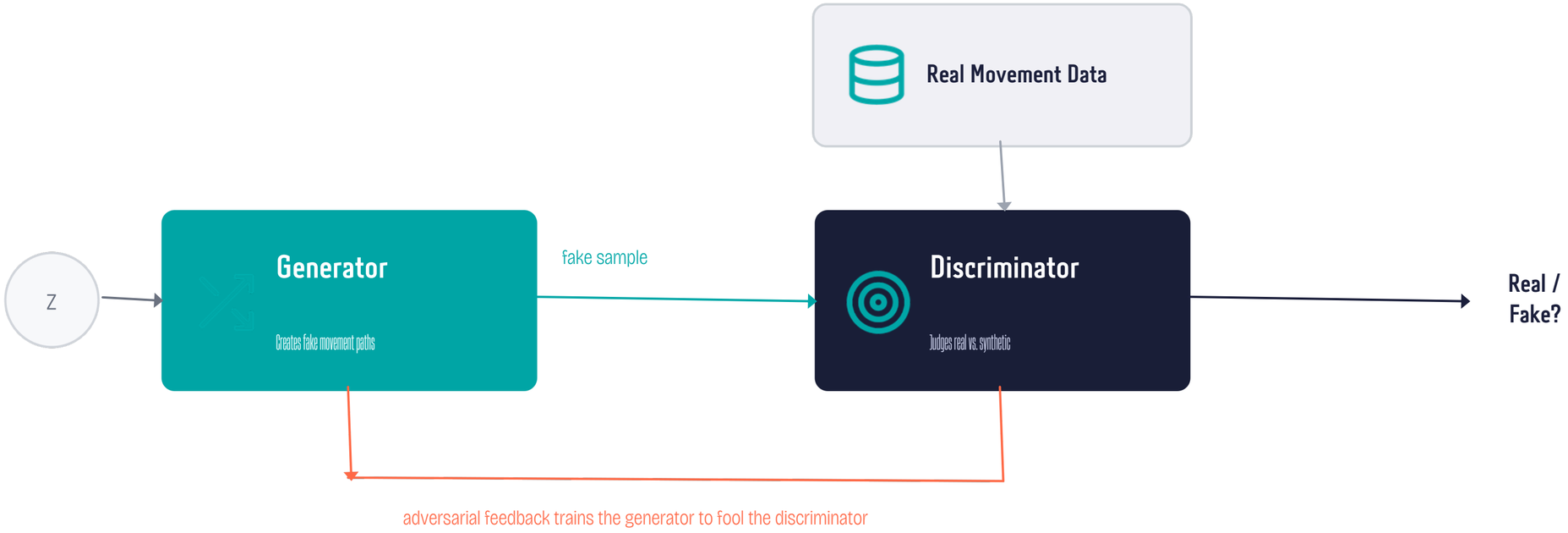
Multi-step-ahead demand forecast used to trigger reorder points automatically.

Impact: accurate short-horizon forecasts reduce stockouts and cut excess inventory holding costs – directly feeding shelf-replenishment decisions.

MODULE 5

GANs: Simulating Customer Movement

GenerativeAdversarialNetworkscreaterealisticsyntheticin-storemovementpathstodensifysparse tracking data.



Densifies sparse trajectories

Augments limited real tracking data with realistic synthetic paths.

Improves recommendations

Richer movement data trains better product-placement & recommendation models.

Pri va cy -fr ie ndl y

Synthetic data reduces reliance on raw, identifiable customer footage.

PUTTING IT TOGETHER

System Integration & Expected Outcomes

Allfivemodelsfeedasingleanalyticsdashboardusedbystoremanagersforstaffing,restocking,andlayout decisions.

↓ 25%

Projected stockout rate
(LSTM-driven reordering)

↑ 2x

Faster model convergence
using Adam vs. SGD

↑ 30%

Moretraining datavia
GAN-augmented trajectories

Rea l-time

Footfall& shelfalerts
viaYOLO detection

Unified Dashboard Signals

- Live foot traffic by zone
- Low-stock shelf alerts
- 7/14/30-day demand forecast
- Customer segment predictions
- Recommendation refresh from synthetic data

CONCLUSION

From Raw Retail Data to Smarter Decisions

MLP

Predicts customer purchasing behavior

SGD vs Adam

Adam chosen for faster convergence

YOLO

Real-time counting & shelf monitoring

LSTM

Forecasts product demand accurately

GAN

Simulates data to boost re comme ndations

Future Scope

- Deploy YOLO on edge devices (in-store cameras) for latency-free alerts
- Extend LSTM to a Transformer-based forecaster for multi-store demand
- Personalize recommendations in real time using GAN-augmented profiles
- Roll out a live manager dashboard with automated reorder triggers

Thank You

Questions & Discussion